

AC-ICD-00108 v1.0

RW-0.01 Interface Control Document

Small Reaction Wheel - 0.01 Nms

Hardware Revision 6

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Approved By:	Systems Engineering

AnyCompany

1. Document History

This section records all revisions made to this document.

Version	Date	Author	Description
1.0	2025-09-15	Engineering Team	Initial release for RW-0.01 hardware revision 6. Full mechanical, electrical, environmental, and software interface definition.

1.1 Applicable Documents

ID	Document	Title
RD-1	AC-DS-00108	RW-0.01 Product Data Sheet
RD-2	AC-UM-00108	RW-0.01 User Manual
RD-3	NASA GSFC-STD-7000A	General Environmental Verification Standard (GEVS)
RD-4	ECSS-Q-ST-70-02C	Thermal Vacuum Outgassing Test
RD-5	RFC 1055	Serial Line Internet Protocol (SLIP)
RD-6	IEC 61000-4-2	Electrostatic Discharge Immunity Test

1.2 Distribution List

Recipient	Organization	Copy Type
Systems Engineering	AnyCompany	Controlled
Flight Software	AnyCompany	Controlled
Quality Assurance	AnyCompany	Controlled
Customer Integration	External	Controlled

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3. Scope and Applicability

This Interface Control Document (ICD) defines the mechanical, electrical, environmental, and software interfaces for the AnyCompany RW-0.01 small reaction wheel, hardware revision 6. It establishes the requirements and constraints that must be satisfied by both the reaction wheel and the host spacecraft to ensure proper integration and operation.

The RW-0.01 is a single-axis reaction wheel providing 0.01 Nms angular momentum storage and 1.0 mNm peak torque. It is designed for ADCS on CubeSats and small satellites (1U to 12U class). The RW-0.01 shares a common design family with the RW-0.03, ensuring high design maturity and parts commonality.

Flight Heritage: 12 units with 105+ cumulative flight years since 2014. Common design with RW-0.03 (90 units, 510+ years since 2008) extends heritage significantly.

Product Classification: Flight-heritage-qualified component. Full qualification testing completed including random vibration, thermal vacuum cycling, radiation TID, and EMC testing.

This document applies to:

- RW-0.01 hardware revision 6
- Firmware version 3.2.x or later
- Production serial numbers SN-0010-0001 and above

4. Mechanical Interface

4.1 Physical Specifications

Parameter	Value	Units	Notes
Mass	<125	g	Including harness pigtail
Dimensions (L x W x H)	50 x 50 x 29.5	mm	Excluding connector protrusion
Mounting Pattern	42 x 42	mm	Square, 4x M3 corner holes
Mounting Fasteners	M3	-	4x SHCS (customer supplied)
Flywheel Inertia	1.5×10^{-5}	kg m ²	About spin axis
Maximum Speed	+/- 6500	RPM	Firmware limited
Center of Mass	Geometric center +/-1	mm	All axes

Figure 1: Mechanical Outline Drawing (Top View)

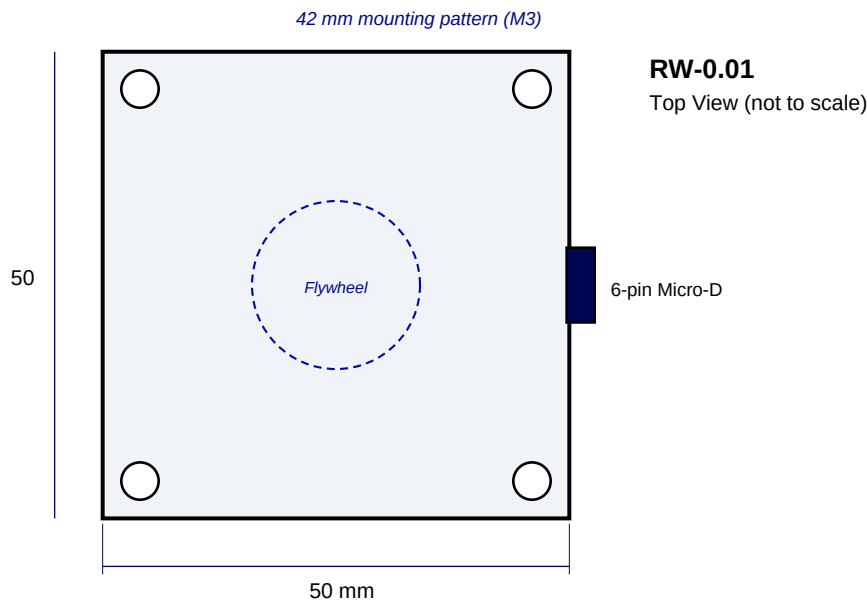


Figure 1: Top-view outline. Square 50x50 mm housing, M3 corner holes (42 mm pattern), 6-pin Micro-D connector, flywheel envelope (dashed).

Figure 2: Side View Profile

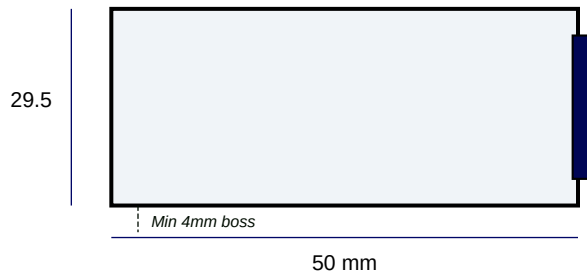


Figure 2: Side profile. 29.5 mm height, minimum 4 mm mounting boss required.

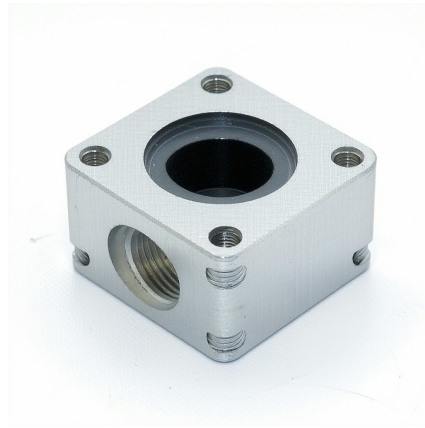


Figure 3: RW-0.01 Product Photograph

4.2 Mounting Interface

Mount using four M3 socket head cap screws through corner holes on a 42 mm square pattern. A minimum 4 mm boss or spacer beneath each point is required to prevent warping. A precision-machined boss is recommended. Torque: 0.5 Nm max. Surface flatness: 0.05 mm across all four points.

Alignment: Spin axis is perpendicular to mounting face within 0.1 degrees. No alignment features; bolt pattern defines orientation.

4.3 Remove Before Flight Items

Item	Action	Notes
Kapton bearing covers	Remove before power-on	2x Kapton tape protecting bearings from debris
Connector dust cap	Remove at integration	Red anodized protective cap. Retain for storage.
Desiccant pack	Remove from bag	Dispose per local procedures.
Shipping restraint	Verify removed	Foam insert must not be installed on spacecraft.

5. Environmental Requirements

5.1 Storage

Store in clean environment (ISO Class 8 or better). Humidity below 60%. For storage >30 days, seal in ESD bag with desiccant. Storage temp: -20C to +50C.

5.2 Thermal

Parameter	Min	Typ	Max	Units	Notes
Survival Temperature	-40	-	+100	C	Non-operating, baseplate
Operating Temperature	-30	+25	+70	C	At mounting interface
Thermal Gradient	-	-	10	C/min	Rate of change limit
Qualification Range	-45	-	+80	C	Derated 10C for flight
Turn-on Temperature	-30	-	+70	C	Safe to apply power
Bearing Grease Range	-40	-	+80	C	Braycote 601EF

Temperature measured at baseplate. Thermal conductance motor-to-baseplate: ~0.8 W/K.

5.3 Pressure and Outgassing

Qualified for sea-level atmosphere and hard vacuum ($<10^{-5}$ Torr). Not qualified for intermediate pressures (Paschen region). Do not power during ascent without additional testing. Materials: TML < 1.0%, CVCM < 0.1% per ECSS-Q-ST-70-02C.

5.4 Vibration

Test	Level	Duration	Axes	Standard
Random Vib. (Acceptance)	12.87 Grms	1 min/axis	3	NASA GEVS
Random Vib. (Qualification)	+3 dB (~15.8 Grms)	2 min/axis	3	NASA GEVS
Sine Sweep	0.5g, 2 oct/min	1 sweep/axis	3	Characterization
Sine Burst	10g @ 20 Hz	5 cycles/axis	3	Quasi-static

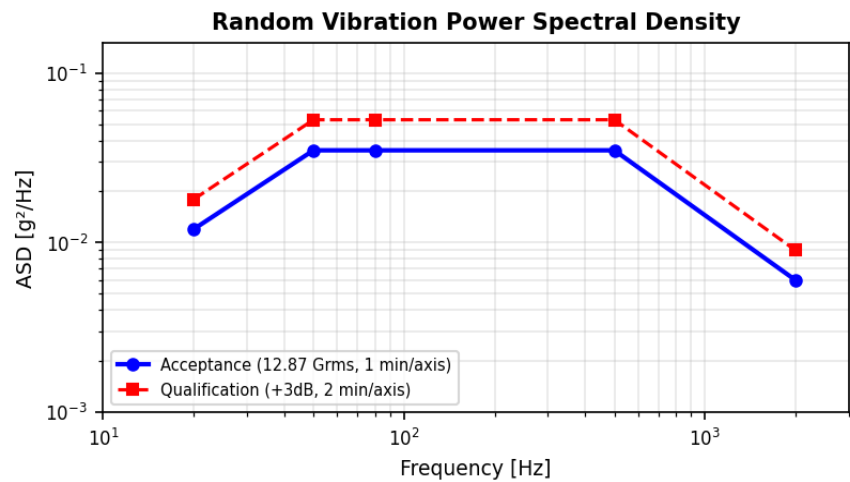


Figure 4: Random Vibration PSD (Acceptance and Qualification)

PSD Breakpoints:

Freq (Hz)	Slope	Accept. ASD (g^2/Hz)	Qual. ASD (g^2/Hz)
20	+6 dB/oct	0.012	0.018
50	flat	0.035	0.053
80	flat	0.035	0.053
500	-6 dB/oct	0.035	0.053
2000	-	0.006	0.009
Overall		12.87 Grms	~15.8 Grms

5.5 Radiation

Electronics: **24 krad** TID at board level, MIL-STD-883 Method 1019, low dose rate. Motor/bearings inherently tolerant.

Parameter	Value	Notes
TID (board level)	24 krad	MIL-STD-883, low dose rate
SEL threshold	>60 MeV cm^2/mg	No destructive latchup
SET rate	<1 per day at GEO	Recoverable via watchdog

6. Electrical Interface

6.1 Power Architecture

Single power input (3.5-6.0V). DC/DC converts to internal 3.3V. H-bridge motor driver with regenerative braking (pushes current back to bus during deceleration). Spacecraft power must accommodate regen current. OVP resets wheel at ~6.5V.

Figure 5: Power Architecture Block Diagram

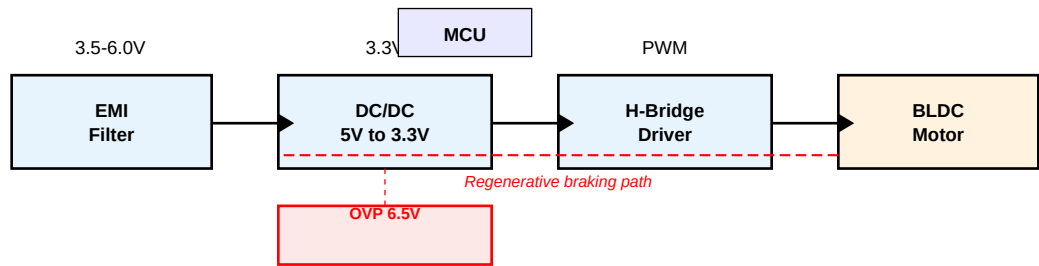


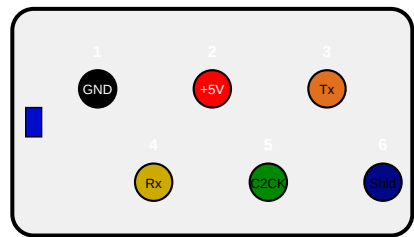
Figure 5: EMI filter, DC/DC, H-bridge motor driver, regenerative path, OVP, and MCU.

6.2 Connector and Pinout

6-contact Micro-D. Mating connector: AnyCompany-MDM-6-F or equivalent. Keyed to prevent mismatch.

Pin	Signal Name	Wire Color	Description	Voltage Level
1	GND	Black	Power return / signal ground	0V reference
2	+5V Power	Red	Primary power input	3.5 - 6.0V
3	Tx	Orange	UART transmit (wheel to host)	3.3V LVTTTL
4	Rx	Yellow	UART receive (host to wheel)	3.3V LVTTTL
5	C2CK	Green	Programming/debug (factory only)	3.3V
6	Shield	Bare/Silver	Chassis/EMI ground	Chassis ref

Figure 6: Connector Face View (Mating Side)



View: Mating face (cable side)

Figure 6: 6-pin Micro-D face view. Key notch at left prevents mismatch.

6.3 Power Specifications

Parameter	Min	Typ	Max	Units	Notes
Operating Voltage	3.5	5.0	6.0	V	At connector pins 1-2
Absolute Max	-0.3	-	7.0	V	Permanent damage above this
Power (Idle)	-	0.05	0.1	W	Motor stationary
Power (Nominal)	-	0.2	-	W	Steady-state, speed dependent
Power (Peak)	-	-	0.8	W	Max torque acceleration
Inrush Current	-	-	0.5	A	Duration < 5 ms
Steady-State Current	-	40	160	mA	At 5V, speed dependent
Front-end Cap	-	-	470	nF	EMI filter
Total Bus Cap	-	-	100	uF	All internal caps
Regen Current	-	-	-50	mA	Source during decel

6.4 Signal Specifications

Parameter	Min	Typ	Max	Units	Notes
Vih (Logic High)	2.3	3.3	3.6	V	3.3V LVTTTL
Vil (Logic Low)	-0.3	0	0.8	V	3.3V LVTTTL
Vol (Output Low)	-	-	0.4	V	At 4 mA sink
Voh (Output High)	2.4	-	-	V	At -2 mA source
ESD (HBM)	-	-	+/-4	kV	IEC 61000-4-2 contact
Baud Rate	-	115200	-	bps	Fixed
Max Cable Length	-	-	3	m	Recommended max

7. Software Interface

7.1 Sign Convention

Positive angular momentum/speed = counter-clockwise (CCW) viewed from top (connector side). Positive torque accelerates CCW.

7.2 Protocol Layer 2 - Physical

Asynchronous UART: 115200 baud, 8N1, 3.3V LVTTTL. Host Tx to wheel Rx (pin 4), host Rx to wheel Tx (pin 3). Always active when powered. No hardware flow control.

Parameter	Value	Notes
Baud Rate	115200 bps	Fixed
Data Bits	8	LSB first
Parity	None	-
Stop Bits	1	-
Flow Control	None	SLIP framing

7.3 Protocol Layer 3 - NSP/SLIP

Nanosatellite Protocol (NSP) with SLIP framing per RFC 1055. Each message wrapped between FEND delimiters. Occurrences of FEND/FESC in body are escaped.

Symbol	Hex	Description
FEND	0xC0	Frame End - start and end delimiter
FESC	0xDB	Frame Escape - precedes escaped byte
TFEND	0xDC	Transposed FEND - literal 0xC0 in payload
TFESC	0xDD	Transposed FESC - literal 0xDB in payload

7.4 Protocol Layer 4 - NSP Message

Figure 7: NSP Message Frame Structure

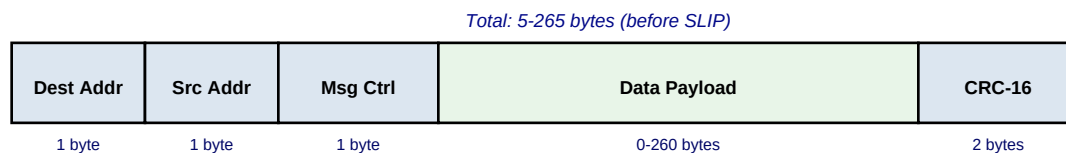


Figure 7: NSP message frame structure.

Byte Order: Little-endian (LSB first).

CRC: CRC-CCITT $x^{16}+x^{12}+x^5+1$, init 0xFFFF, LSB first.

Msg Control: Bit7=P/F, Bit6=B, Bit5=ACK, Bits4-0=Command Code.

Max Size: 265 bytes raw, 532 after SLIP (worst case).

7.5 Protocol Layer 5 - Commands

Code	Name	Direction	Description	Data Len
0x00	PING	CMD/REPLY	Heartbeat, resets watchdog	0 / 0
0x01	INIT	CMD	Mode transition (boot/app)	0 or 4
0x02	PEEK	CMD/REPLY	Read memory address	4 / var
0x03	POKE	CMD/REPLY	Write memory address	var / 0
0x04	DIAGNOSTIC	CMD/REPLY	Read ADC channel	1 / 2
0x06	CRC	CMD/REPLY	Compute firmware CRC	8 / 2
0x07	READ_FILE	CMD/REPLY	Read file system	4 / var
0x08	WRITE_FILE	CMD/REPLY	Write file system	var / 0
0x09	READ_EDAC	CMD/REPLY	Read EDAC memory	4 / var
0x0A	SET_SPEED	CMD/REPLY	Set speed (RPM, int16)	2 / 0
0x0B	SET_TORQUE	CMD/REPLY	Set torque (mNm x10, int16)	2 / 0
0x0C	GET_TELEMETRY	CMD/REPLY	Request telemetry packet	0 / 14

7.6 Protocol Layer 6 - Application

Mode	ID	Description	Transitions From
Idle	0	Motor off, telem active, accepts all cmds	Any (auto on fault)
Speed Control	1	Closed-loop PID to RPM setpoint	Idle (SET_SPEED)
Torque Control	2	Open-loop torque command	Idle (SET_TORQUE)
Test	3	Factory diagnostics (not for flight)	Idle (POKE mode reg)

Watchdog: Host must send valid command within 500 ms. On timeout: auto-Idle, sets watchdog_reset bit. Resume on next valid command. Timeout configurable via register 0x0024.

8. Telemetry Definitions

GET_TELEMETRY (0x0C) returns 14 bytes:

Offset	Field	Type	Units	Range / Notes
0-1	wheel_speed	int16	RPM	-6500 to +6500 (CCW positive)
2-3	motor_current	uint16	mA	0-500 (RMS phase current)
4-5	bus_voltage	uint16	mV	0-8000 (at input)
6	pcb_temperature	int8	deg C	-40 to +100
7-8	fault_state	uint16	bitfield	See Section 10
9	hall_state	uint8	bitfield	Bits 2:0 = H1:H2:H3
10-13	uptime	uint32	seconds	Since last reset

8.1 Diagnostic Channels

DIAGNOSTIC (0x04) reads individual ADC channels. Data byte = channel ID. Reply = 12-bit ADC (uint16).

Ch	Description	Units	Resolution	Scaling
0	Wheel speed	RPM	1 RPM	From hall timing
1	Motor current (RMS)	mA	0.5 mA	0.488 mA/count
2	Bus voltage	mV	1 mV	1.953 mV/count
3	PCB temperature	deg C	0.5 C	NTC lookup
4	3.3V reference	mV	1 mV	1.953 mV/count
5	Motor back-EMF	mV	1 mV	Dead-time measurement

9. Register and Memory Map

EDAC memory: 256 bytes, 0x0000-0x00FF. Access via PEEK (0x02), POKE (0x03), READ_EDAC (0x09). R/W writes take effect next control loop (~1 ms).

Address	Field	Type	R/W	Description
0x0000-0x0001	speed_setpoint	int16	R/W	Commanded speed (RPM)
0x0002-0x0003	torque_setpoint	int16	R/W	Commanded torque (mNm x10)
0x0004-0x0005	speed_measure	int16	R	Current speed (RPM)
0x0006-0x0007	motor_current	uint16	R	Motor current (mA)
0x0008-0x0009	bus_voltage	uint16	R	Bus voltage (mV)
0x000A	temperature	int8	R	PCB temp (deg C)
0x000B-0x000C	fault_state	uint16	R/W	Fault bitfield (write 0 to clear)
0x000D	hall_state	uint8	R	Hall sensor state
0x000E	mode	uint8	R/W	0=Idle 1=Speed 2=Torque 3=Test
0x000F	watchdog_count	uint8	R	Timeout counter
0x0010-0x0013	cumulative_revs	uint32	R	Lifetime revolutions
0x0014-0x0017	operating_hours	uint32	R	Powered seconds
0x0018-0x0019	fw_version	uint16	R	Major.Minor (0x0302=3.2)
0x001A-0x001B	serial_number	uint16	R	Factory serial
0x001C-0x001D	speed_kp	int16	R/W	Speed P gain (x100)
0x001E-0x001F	speed_ki	int16	R/W	Speed I gain (x100)
0x0020-0x0021	torque_limit	uint16	R/W	Max torque (mNm x10)
0x0022-0x0023	speed_limit	uint16	R/W	Max speed (RPM)
0x0024	watchdog_timeout	uint8	R/W	100ms increments (def 5=500ms)
0x0025	node_address	uint8	R/W	NSP node addr (def 0x10)
0x0026-0x00FF	reserved	-	-	Reserved (reads 0x00)

10. Fault Handling

fault_state (0x000B-0x000C): 16-bit latching bitfield. Faults latched on detection; clear by writing 0x0000. Critical faults (bits 0-6) force Idle mode.

Bit	Name	Severity	Description	Recovery
0	overcurrent	Critical	Motor current > 500 mA	Auto-idle. Clear to resume.
1	overvoltage	Critical	Bus > 6.5V (OVP triggered)	Wheel resets. Power cycle.
2	undervoltage	Critical	Bus < 3.2V	Auto-idle. Restore voltage.
3	overtemperature	Critical	PCB > +75 C	Auto-idle. Cool, clear.
4	hall_sensor_fault	Critical	Invalid hall state	Auto-idle. Check wiring.
5	comm_timeout	Critical	No cmd within watchdog	Auto-idle. Send PING.
6	motor_stall	Critical	Speed not reached in 5s	Auto-idle. Check obstruction.
7	watchdog_reset	Info	Watchdog expired	Counter incremented.
8-15	reserved	-	Reserved	-

10.1 Fault Recovery Sequence

1. Read fault_state (PEEK 0x000B, 2 bytes) to identify faults.
2. Address root cause (restore voltage, cool down, clear obstruction, re-establish comms).
3. Write 0x0000 to fault_state (POKE 0x000B) to clear latched faults.
4. Issue SET_SPEED or SET_TORQUE to resume desired mode.
5. Read fault_state again to verify 0x0000.

11. Command Examples

Hex byte sequences for common operations. SLIP framing (FEND=0xC0) wraps each message. Default wheel address: 0x10, host: 0x01.

11.1 PING

Purpose: Verify link, reset watchdog.

Command (host to wheel):

```
C0 10 01 00 [CRC_L] [CRC_H] C0
```

0x10=wheel addr, 0x01=host addr, 0x00=PING cmd. CRC over 3 message bytes.

Reply (wheel to host):

```
C0 01 10 20 [CRC_L] [CRC_H] C0
```

0x20 = ACK bit set (bit5) + PING code (0x00).

11.2 SET_SPEED (1000 RPM CCW)

```
C0 10 01 0A E8 03 [CRC_L] [CRC_H] C0
```

0x0A=SET_SPEED, 0x03E8=1000 (little-endian: E8 03). Wheel ramps to 1000 RPM CCW using internal PID.

11.3 GET_TELEMETRY

Command:

```
C0 10 01 0C [CRC_L] [CRC_H] C0
```

Reply (14 data bytes):

```
C0 01 10 2C E8 03 1E 00 88 13 19 00 00 00 64 00 00 00 [CRC_L] [CRC_H] C0
```

Decoded: speed=0x03E8=1000 RPM, current=0x001E=30 mA, voltage=0x1388=5000 mV, temp=0x19=25C, fault=0x0000, hall=0x00, uptime=0x00000064=100 sec.

11.4 SET_TORQUE (0.5 mNm)

```
C0 10 01 0B 05 00 [CRC_L] [CRC_H] C0
```

0x0B=SET_TORQUE, 0x0005=5 (= 0.5 mNm, since units are mNm x10). Positive = CCW torque.

11.5 PEEK (Read Speed Setpoint Register)

```
C0 10 01 02 00 00 00 00 [CRC_L] [CRC_H] C0
```

0x02=PEEK, address=0x00000000 (speed_setpoint). Reply includes 2-byte value at that address.

12. Integration Checklist

Complete each step and record date/initials in your integration log.

#	Step	Category	Verification
1	Remove all RBF items	Mechanical	Visual inspection
2	Verify boss height ≥ 4 mm (4 points)	Mechanical	Calipers measurement
3	Install M3 screws, torque to 0.5 Nm	Mechanical	Torque wrench
4	Verify flatness < 0.05 mm	Mechanical	Surface gauge
5	Mate 6-pin Micro-D, verify pin alignment	Electrical	Continuity test
6	Apply power (5V), verify < 100 mA idle	Electrical	Multimeter
7	Send PING, verify reply	Software	Terminal/script
8	Read firmware version (0x0018)	Software	PEEK command
9	Command +100 RPM, verify telem matches	Functional	GET_TELEMETRY
10	Command -100 RPM, verify sign convention	Functional	GET_TELEMETRY
11	Test watchdog: stop cmds for 600 ms	Functional	Observe fault bit 7
12	Read all diagnostic channels (0-5)	Functional	DIAGNOSTIC cmd
13	Full speed sweep +/-6500 RPM	Functional	Automated script
14	Verify regen: decel, monitor bus voltage	Electrical	Oscilloscope
15	Record SN, FW version, cal date	Documentation	PEEK commands

Appendix A: Acronyms and Abbreviations

Acronym	Definition
ADCS	Attitude Determination and Control Subsystem
ADC	Analog-to-Digital Converter
BLDC	Brushless Direct Current
CCW	Counter-Clockwise
CRC	Cyclic Redundancy Check
CVCM	Collected Volatile Condensable Material
DC/DC	DC-to-DC Converter
EDAC	Error Detection and Correction
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
ESD	Electrostatic Discharge
GEVS	General Environmental Verification Standard
HBM	Human Body Model
ICD	Interface Control Document
LVTTTL	Low-Voltage TTL
MCU	Microcontroller Unit
NSP	Nanosatellite Protocol
OVP	Overvoltage Protection
PID	Proportional-Integral-Derivative
PSD	Power Spectral Density
RBF	Remove Before Flight
RPM	Revolutions Per Minute
SEE	Single Event Effects
SEL	Single Event Latchup
SET	Single Event Transient
SHCS	Socket Head Cap Screw
SLIP	Serial Line Internet Protocol
TID	Total Ionizing Dose
TML	Total Mass Loss
UART	Universal Asynchronous Receiver-Transmitter

End of Document

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